

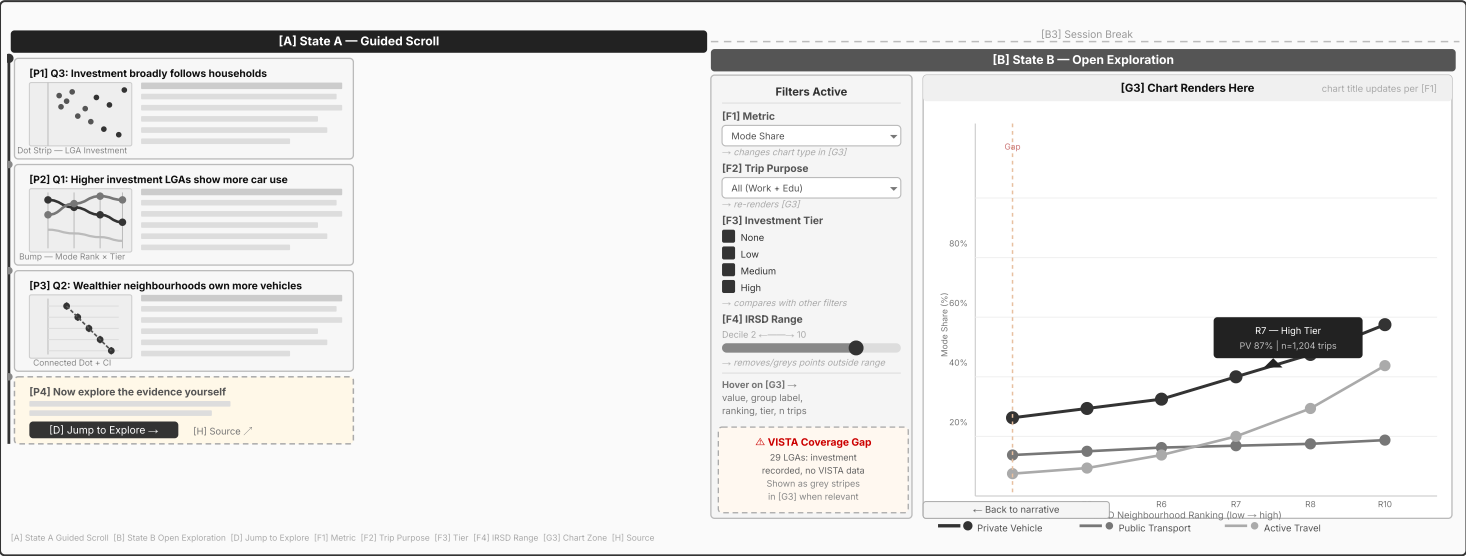
INFORMATION

Title: Design C — Martini Glass | **Sheet No:** Sheet #4
Author: Bharath Arun Gandhimani | **Date:** 13/05/2026
Description: Interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity and household travel behaviour in Victoria to policymakers and transport planners. A two-state structure: State A delivers a guided scroll through four sequential narrative panels [P1–P4], each pairing a bold headline claim with a small embedded chart. A [D] button transitions the reader to State B, where filters [F1–F4] drive a live chart zone [G3] for open exploration.

DESIGN RATIONALE

The martini glass structure gives policymakers the narrative framing they need before exploration — author controls the story entry, reader controls the exit. Chart types are chosen to be fully distinct from Sheets 2 and 3 in both chart family AND analytical task — geographic/categorical (Sheet 2) vs distributional/bivariate (Sheet 3) vs ranking/trend/paired (Sheet 4): **dot strip plot** (investment dispersion across LGAs, Q3), **bump/slope chart** (mode share rank shift across investment tiers, Q1 — redesigned as a slope/connected line chart in Sheet 5 for legibility across four discrete tiers), and **connected dot plot with 95% CI error bars** (vehicle ownership by income band, Q2). [F1] Metric dropdown switches the chart type in [G3] — a mode share metric renders a connected dot plot, a duration metric renders a violin+box plot, a vehicle ownership metric renders a connected dot plot with CI bars.

LAYOUT



OPERATIONS

[A] Scroll / natural read through [P1–P3]

⇒

Reader moves through three authored panels in sequence. Each panel is a fixed layout: chart thumbnail (left) + headline claim + narrative text (right). No interaction required — author controls the evidence order. [P4] bridge panel appears at the bottom of the scroll.

[D] Click "Jump to Explore"

⇒

Transitions from State A to State B. Left narrative column [P1–P3] is replaced by the filter panel [F1–F4]. Filters activate from greyed state. [G3] renders with default settings: F1 = Mode Share, F2 = All, F3 = All tiers checked, F4 = full IRSD range. A "← Back to narrative" link stays visible.

[F1] Change Metric dropdown

⇒

[G3] chart type changes: Mode Share → connected dot plot; Trip Duration → violin + box plot; Vehicle Ownership → connected dot plot with 95% CI error bars. Chart title in [G3] updates accordingly. All other active filters [F2–F4] remain applied.

[F2] Change Trip Purpose dropdown

⇒

[G3] re-renders for selected purpose subset (Work / Education / Both). Compares with [F1] and [F3] simultaneously. Chart title updates to reflect selected purpose.

[F3] Change Investment Tier checkboxes

⇒

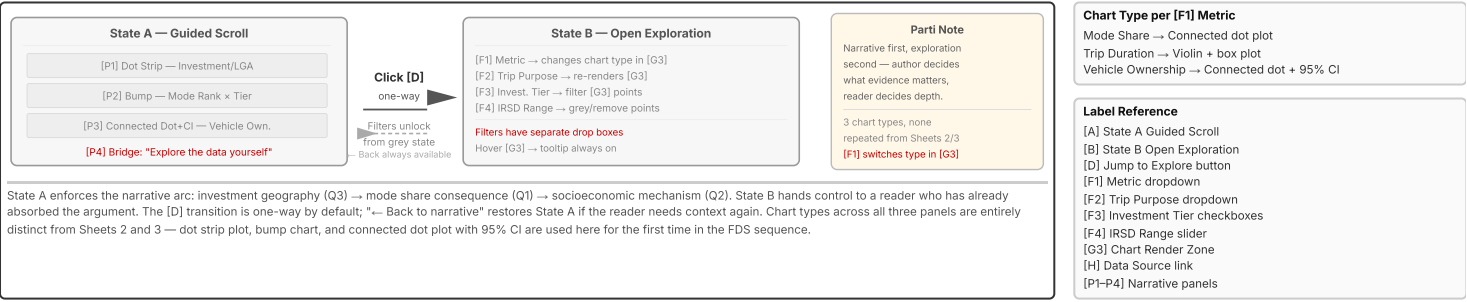
[G3] re-renders for selected tier(s) only. Unchecked tiers are removed from view. Chart title updates. Compounds with other active filters [F1] and [F2].

[F4] Drag IRSD Range slider

⇒

Data points outside selected IRSD decile range are removed or greyed out in [G3]. Compounds with other filters. Hover on any point in [G3] → tooltip: value, group label, ranking, tier, n trips. [H] data source link always accessible.

FOCUS / PART I — TWO-STATE TRANSITION



DISCUSSION

Pros

- Policymakers receive a structured evidence briefing before controls are handed over — they understand the argument before they can challenge it.
- Small, paired chart-text accumulation in [P1–P3] is cognitively natural: one claim, one chart, one reading at a time.
- [D] gives immediate access to experienced policymakers wanting to explore the data without sitting through the full narrative.
- [P4] makes it clear to readers that they have finished reading and are entering a different mode — reducing the ambiguity of the transition.
- "← Back to narrative" link enables users to come back to the guided mode without losing exploration state.
- Three chart types in [P1–P3] — dot strip plot, bump chart, and connected dot plot — each belong to a different visual grammar family (strip/dispersion, ranking/trajectory, paired comparison). None appear in Sheets 2 or 3. This ensures the three alternative designs address different perceptual tasks and satisfy the FDS requirement for structurally distinct alternatives.

Cons

- For experienced policymakers, the static narrative guided mode may feel slow — they may jump to [D] immediately, bypassing the intended framing.
- Placing the exploration zone on the right side in State B may make the reader think there is an exploration mode before even reading the guided version.
- Four stacked narrative panels make the left column long — readers on small screens face significant scrolling before reaching the [D] bridge panel and exploration mode.
- [F1] switching the chart type in [G3] changes the visual encoding significantly — a reader who has not read the panel descriptions may be confused by the sudden appearance of a violin plot rather than a dot plot.
- The four-filter exploration zone [F1–F4] risks structural convergence with Design B's scoped-filter dashboard — both designs use multiple filter controls to drive chart output. This reduces the interaction-model distinctness that the FDS requires across Sheets 2–4. Addressed in Sheet 5 by replacing the filter zone with a single LGA personalisation dropdown, restoring genuine interaction distinctness.